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Sub ject : DSA

Unit 3, S6, SLO2 - Stack Applications - Tower of Hanoi

1. ____ is a classic problem that demonstrates the use of recursion with stacks.

Answer: Tower of Hanoi

2. The minimum number of moves required to solve the Towers of Hanoi with n disks is ____.

Answer: $2^n - 1$

3. The recursive solution to Towers of Hanoi uses the ____ data structure to manage function calls.

Answer: call stack

4. In the Towers of Hanoi, disks are moved from the source rod to the ____ rod using an auxiliary rod.

Answer: destination (target)

5. Each recursive call pushes a new ____ onto the stack.

Answer: activation frame (state)

6. The Towers of Hanoi involves ____ recursive steps.

Answer: two ($T(n-1)$ moves)

7. The stack unwinds when the ____ case of the recursive function is reached.

Answer: base ($n == 1$)

8. The base condition in Towers of Hanoi is when ____ disk is to be moved.

Answer: one (single)

9. Stacks in recursion help to ____ the order of operations in solving the puzzle.

Answer: track and restore

10. The auxiliary rod in the Towers of Hanoi acts as a ____ during disk transfers.

Answer: temporary buffer (helper)